

Oncology

An ounce of prevention

More than a third of cancers arise from preventable risks

SCIENTISTS HAVE long known that some cancers have preventable causes. But reliable estimates of how many, exactly, have been few and far between. In a report published in *Nature Medicine* on February 3rd, a team led by researchers at the International Agency for Research on Cancer at the World Health Organisation provide the most comprehensive figures to date.

The 30 risk factors covered in the study include habits within an individual's control, such as smoking and alcohol consumption, as well as environmental factors such as air pollution and infections. The study found that of the nearly 20m new cancers thought to have occurred worldwide in 2022 (the last year with available data), 38% were due to preventable factors. Because the study did not include the effects of various suspected carcinogens, such as certain food preservatives, the authors estimate the real figure may be somewhat higher. For now, the study offers policymakers their clearest guide yet on how public-health measures can reduce the rates of cancer.

Unsurprisingly, people living in different parts of the world are exposed to different risks (see chart on next page). But two stand out virtually everywhere: tobacco smoking and infections. Smoking is the leading cause of cancers in men in almost all countries outside sub-Saharan Africa, as well as for women in America, Europe and Oceania. Infections are the leading cause for women elsewhere. All told, one in six cancers worldwide are caused by smoking and one in ten is caused by an infection. Alcohol, third in the overall ranking, causes 4.6% of all cancers in men and 1.6% of those in women.

The results highlight the continued health risks posed by smoking, which has been linked to at least 15 types of cancer. Even those who quit can have a heightened risk of developing cancer for decades afterwards. Many countries are, therefore, trying to catch lung cancer earlier by introducing routine CT scans for both current and former smokers.

Carcinogenic infections are another area where intervention could do tremendous good. Nearly all cervical cancers, for example, are caused by chronic infection with the human papillomavirus, or HPV. Liver cancers arise predominantly from the Hepatitis B and C viruses. Stomach cancers are mostly caused by infection with

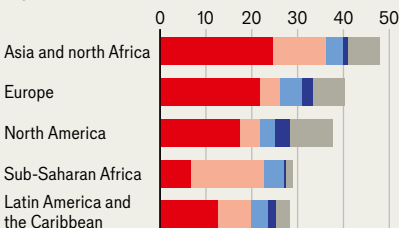


Living dangerously

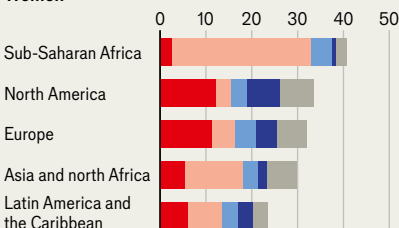
Cancers from preventable causes, 2022, % of all cases

Smoking Infections Alcohol
High BMI Other

Men



Women



Source: "Global and regional cancer burden attributable to modifiable risk factors to inform prevention", by H. Fink, O. Langselius, J. Vignat et al., *Nature Medicine* 2026

► the bacterium *Helicobacter pylori*.

Vaccines against HPV and Hepatitis B infections, now part of many countries' routine jabs for children, are consequently forecast to prevent millions of cancers in the coming years. In Britain, cervical-cancer rates among women in their 20s have fallen by 90% since 2008, when the country began using the HPV jab.

Although there is no vaccine for Hepatitis C, an infection transmitted via blood, highly effective antiviral treatments have become available in the past ten years. Improvements in food hygiene, sanitation and antibiotic availability over the past century, meanwhile, have caused *H. pylori* infection numbers to plummet. These, too, exhibit great regional variability: early-life infection with *H. pylori* remains widespread in poorer countries, where stomach cancer is much more common.

Although there is cause for optimism, some types of cancer remain far less amenable to prevention than others. The onset of cancers of the breast and pancreas, for instance, is mostly due to internal biological mechanisms that scientists have yet to untangle. The hope remains, however, that these, too, may one day join the ranks of cancers with mostly preventable causes. ■